

Sales Performance Analysis

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Problem Statement of Sales Performance Analysis

The purpose of this project is to analyse sales data to understand the factors that affect sales and profit. The dataset contains information about products, units sold, sales, profit, discounts, customers, segments, countries, and time periods. The analysis will identify high-performing products, sales and profit trends, and relationships between important variables. The findings will help identify areas for improving sales performance, profitability, and business decision-making

Project Description of the Sales Performance Analysis

This project analyses sales transaction data to uncover key drivers of revenue, profitability, and discount behaviour across a global business. The dataset contains 700 records spanning 2013–2014, covering 5 customer segments (Government, Enterprise, Midmarket, Small Business, and Channel Partners), 5 countries (United States, Canada, Germany, France, and Mexico), and 6 products (Carretera, Paseo, VTT, Velo, Amarilla, and Montana).

Each record captures units sold, manufacturing price, sale price, gross sales, discounts applied (categorized as None, Low, Medium, or High), net sales, cost of goods sold (COGS), and resulting profit, along with the transaction date.

The goal of the analysis is to evaluate sales and profitability trends across segments, countries, and products over time, assess how discounting impacts net sales and margins, and identify top- and under-performing areas of the business to support data-driven decisions on pricing, discounting strategy, and market/segment focus.

End Users of the Project

Sales Managers / Sales Directors – to track team and regional performance, monitor targets, and identify top-performing products or segments.

Business/Data Analysts – to explore trends, build reports, and generate insights from the raw sales data.

Finance Team – to review profitability, COGS, and margin impact, especially in relation to discounting.

Marketing Team – to understand which segments, countries, or products respond best to promotions/discounts, guiding campaign strategy.

Product Managers – to assess how individual products (Carretera, Paseo, VTT, Velo, Amarilla, Montana) are performing across markets.

Regional/Country Managers – to compare performance across the US, Canada, Germany, France, and Mexico and adjust local strategy.

Executive Leadership (C-suite) – to get a high-level view of overall revenue, profit trends, and strategic decision-making inputs (e.g., which segment/country to invest in).

Pricing/Discount Strategy Team – to evaluate how different discount bands (None, Low, Medium, High) affect net sales and profitability, and optimize discount policy.

Technologies Used In this Project

Hardware Requirements

The hardware requirements for performing the Sales Performance Analysis are:

RAM – 8.00 GB

Operating System – Windows 11

Processor – Intel(R) Core(TM) i3-1115g4

Processor speed – 3.00 GHz

Software Requirements

The software requirements for performing the Sales Performance Analysis are:

Programming Language – Python

Development Environment / IDE – Jupyter Notebook

Data Analysis Libraries – Pandas, NumPy

Data Visualization Libraries – Matplotlib, Seaborn

Dataset Format – CSV

Exploratory Data Analysis (EDA)

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a vital phase in data analysis involving the examination and visualization of data to uncover patterns, trends, and insights. In the context of an Airbnb data analysis project, EDA allows researchers to gain a comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary statistics, data visualization, and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, guiding subsequent analytical decisions.

EDA Questions and Answers

Which segment has the highest sales?

Which country has the lowest sales?

Which Product has the Highest Sales

Which product has highest in the profit

Which Year has highest in the Sales

Which product has the highest units sold?

What are the major sales and profit trends in the dataset?

Are there any outliers in Sales, Profit, or Units Sold?

What is the correlation between Units Sold, Sales, COGS, and Profit?

Which segment has the highest sales?

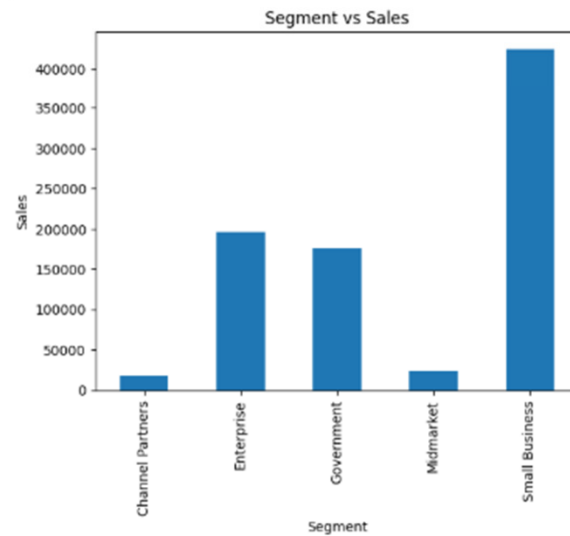
```
[116]: segment = df.groupby("Segment")["Sales"].mean()
print("Highest Sales of Segment", segment)

Highest Sales of Segment Segment
Channel Partners    18005.940000
Enterprise          196117.050000
Government          175014.276667
Midmarket           23818.830000
Small Business     424279.200000
Name: Sales, dtype: float64

[117]: segment = df.groupby("Segment")["Sales"].mean().idxmax()
print("Highest Sales Segment", segment)

Highest Sales of Segment Small Business

[118]: segment = df.groupby("Segment")["Sales"].mean().plot.bar()
plt.title("Segment vs Sales")
plt.xlabel("Segment")
plt.ylabel("Sales")
plt.show()
```



Which country has the lowest sales?

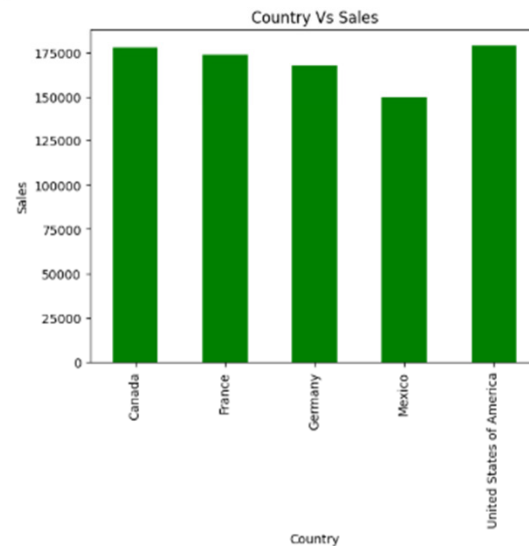
```
[114]: country = df.groupby("Country")["Sales"].mean()
print("Lowest Sales of Country", country)

Lowest Sales of Country Country
Canada                177769.000000
France                173958.442857
Germany               167895.371429
Mexico                149638.264286
United States of America 178784.528571
Name: Sales, dtype: float64
```

```
[115]: country = df.groupby("Country")["Sales"].mean().idxmin()
print("Lowest Sales of Country", country)

Lowest Sales of Country Mexico
```

```
[118]: country = df.groupby("Country")["Sales"].mean().plot.bar(color="green")
plt.title("Country Vs Sales")
plt.xlabel("Country")
plt.ylabel("Sales")
plt.show()
```



Which Product has the Highest Sales

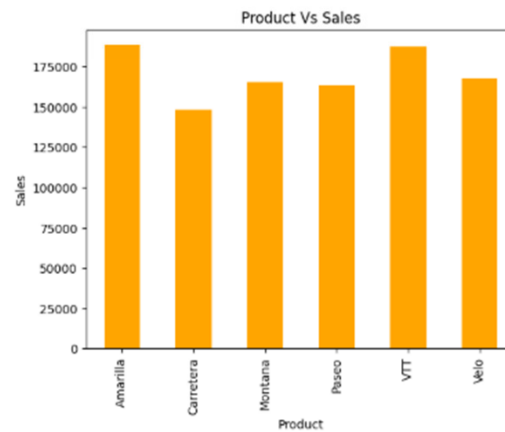
```
[118]: product = df.groupby("Product")["Sales"].mean()
print("Product with highest sales",product)

Product with highest sales Product
Amarilla    188799.127668
Carretera   148551.752688
Montana     165492.548387
Pasco       163421.549585
VTT         188182.825688
Velo        167431.798165
Name: Sales, dtype: float64

[119]: product = df.groupby("Product")["Sales"].mean().idxmax()
print("Product with highest sales",product)

Product with highest sales Amarilla

[120]: product = df.groupby("Product")["Sales"].mean().plot.bar(color="Orange")
plt.title("Product Vs Sales")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.show()
```



Which product has highest in the profit

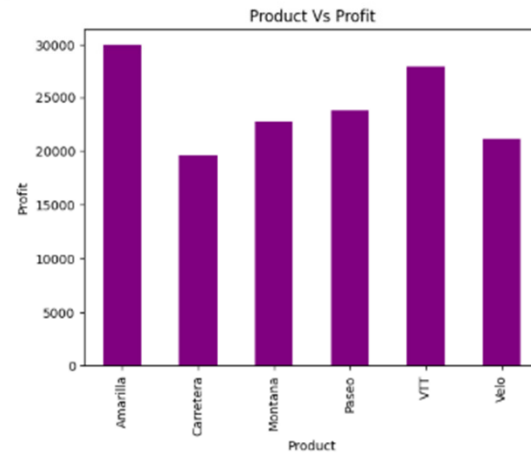
```
[77]: profit = df.groupby("Product")["Profit"].mean()
      print("Product with Highest Profit", profit)

Highest Number of Profit in the Product Product
Amarilla      29937.277234
Carretera     19643.063288
Montana       22739.299785
Paseo         23749.692822
VTT           27840.448558
Velo          21155.894174
Name: Profit, dtype: float64
```

```
[121]: profit = df.groupby("Product")["Profit"].mean().idxmax()
       print("Product with Highest Profit", profit)

Product with Highest Profit Amarilla
```

```
[136]: profit = df.groupby("Product")["Profit"].mean().plot.bar(color="purple")
       plt.title("Product Vs Profit")
       plt.xlabel("Product")
       plt.ylabel("Profit")
       plt.show()
```



Which Year has highest in the Sales

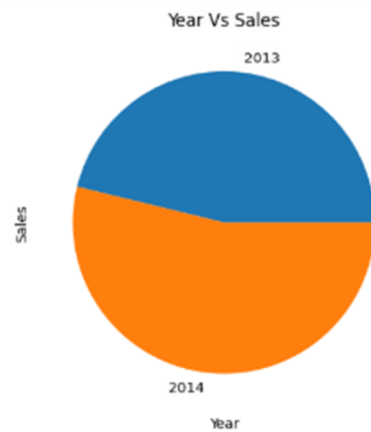
```
[123]: year = df.groupby("Year")["Sales"].mean()
print("Year with highest sales", year)

Year with highest sales Year
2013    150944.388571
2014    175830.699048
Name: Sales, dtype: float64

[124]: year = df.groupby("Year")["Sales"].mean().idxmax()
print("Year with highest sales", year)

Year with highest sales 2014

[137]: year = df.groupby("Year")["Sales"].mean().plot.pie()
plt.title("Year Vs Sales")
plt.xlabel("Year")
plt.ylabel("Sales")
plt.show()
```



Which product has the highest units sold?

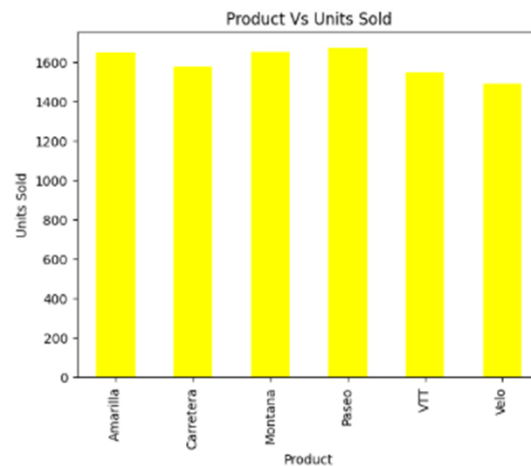
```
[125]: Units_sold = df.groupby("Product")["Units Sold"].mean()
print("Highest number of units sold:", Units_sold)

Highest number of units sold: Product
Amarilla    1652.287234
Carretera   1578.989247
Montana     1658.043011
Pasco       1674.452970
VTT         1548.467890
Velo        1490.133028
Name: Units Sold, dtype: float64

[126]: Units_sold = df.groupby("Product")["Units Sold"].mean().idxmax()
print("Highest number of units sold:", Units_sold)

Highest number of units sold: Pasco

[129]: Units_sold = df.groupby("Product")["Units Sold"].mean().plot.bar(color="yellow")
plt.title("Product Vs Units Sold")
plt.xlabel("Product")
plt.ylabel("Units Sold")
plt.show()
```



What are the major sales and profit trends in the dataset?

```
[92]: country = df.groupby("Country")[["Sales", "Profit"]].sum()
print(country)
```

	Sales	Profit
Country		
Canada	24887660	3529228.885
France	24354182	3781820.780
Germany	23505352	3680388.820
Mexico	20949357	2907523.110
United States of America	25029834	2995540.665

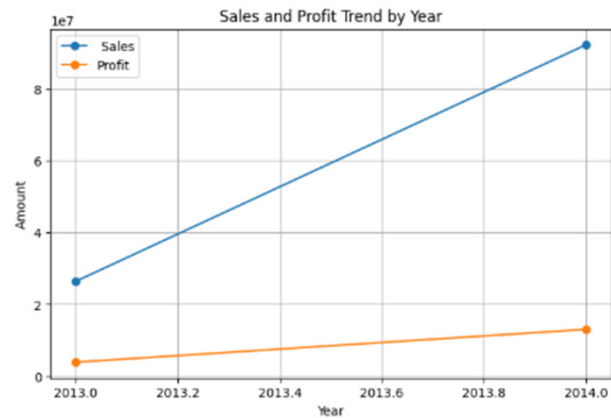
```
[96]: year = df.groupby("Year")[["Sales", "Profit"]].sum()
print(year)
```

	Sales	Profit
Year		
2013	26415268	3878464.51
2014	92311117	13015237.75

```
[108]: import matplotlib.pyplot as plt
```

```
year.plot(kind="line", marker="o", figsize=(8, 5))
```

```
plt.title("Sales and Profit Trend by Year")
plt.xlabel("Year")
plt.ylabel("Amount")
plt.grid(True)
plt.show()
```



```
[97]: segment = df.groupby("Segment")[["Sales", "Profit"]].sum()
print(segment)
```

	Sales	Profit
Segment		
Channel Partners	1800594	1.316803e+06
Enterprise	19611705	-6.145456e+05
Government	52504283	1.138817e+07
Midmarket	2381883	6.601031e+05
Small Business	42427920	4.143168e+06

```
[98]: product = df.groupby("Product")[["Sales", "Profit"]].sum()
print(product)
```

	Sales	Profit
Product		
Amarilla	17747118	2814104.060
Carretera	13815313	1826804.885
Montana	15390807	2114754.880
Paseo	33011153	4797437.950
VTT	20511928	3034608.020
Velo	18250066	2305992.465

Are there any outliers in Sales, Profit, or Units Sold?

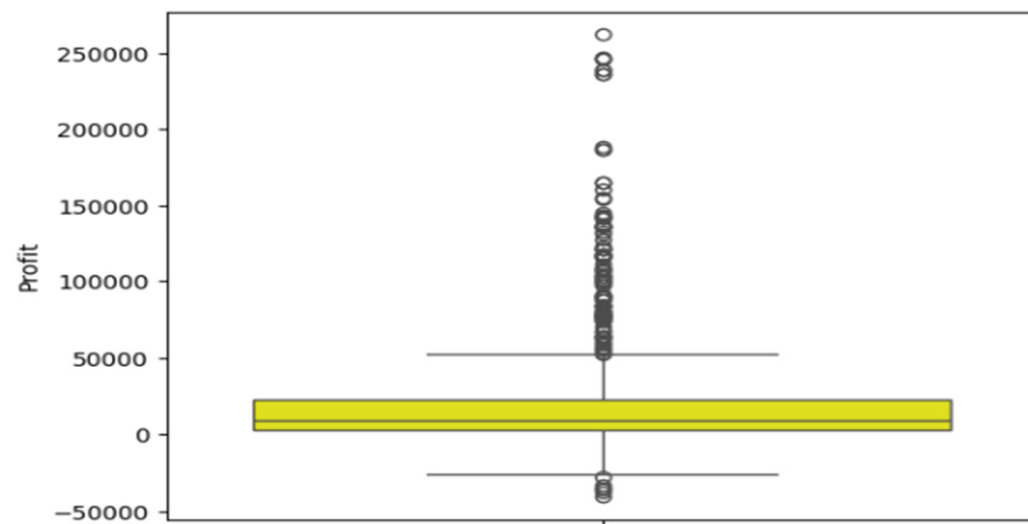
```
[145]: sns.boxplot(y=" Sales" ,data=df,color="Orange")
```

```
[145]: <Axes: ylabel=' Sales'>
```



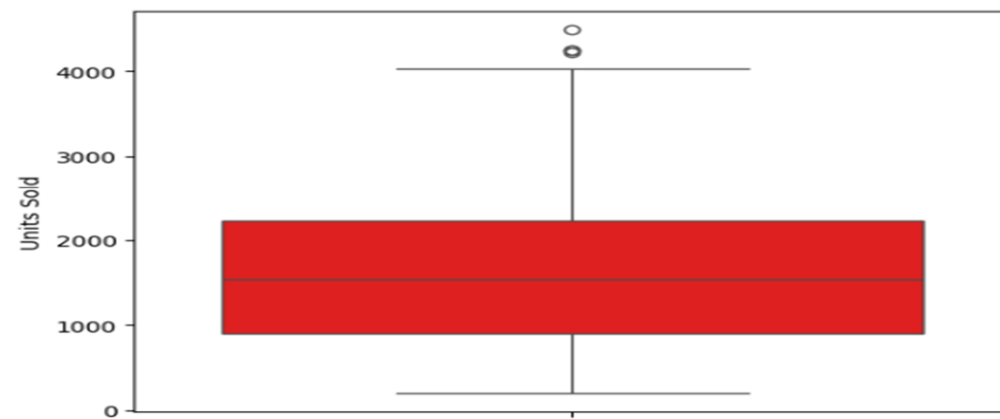
```
[146]: sns.boxplot(y="Profit",data=df,color="yellow")
```

```
[146]: <Axes: ylabel='Profit'>
```



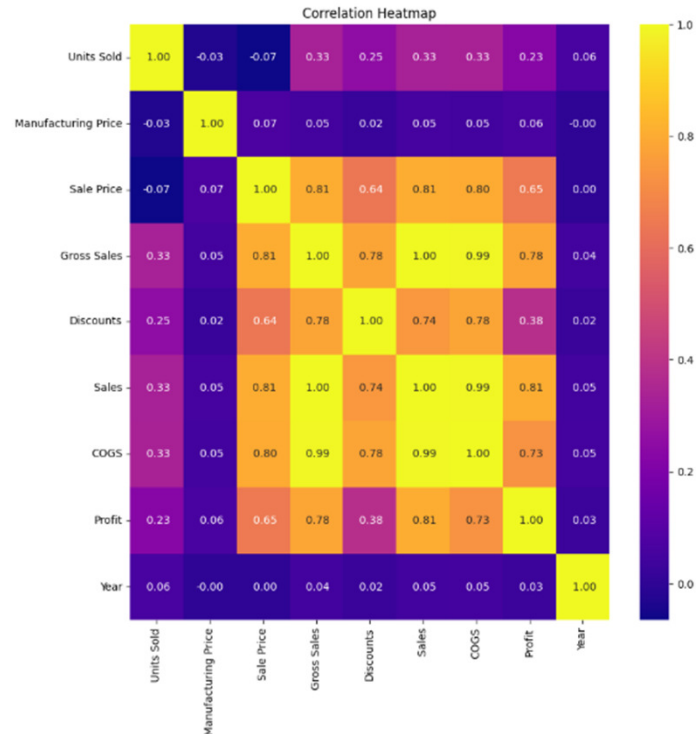
```
[147]: sns.boxplot(y="Units Sold",data=df,color="red")
```

```
[147]: <Axes: ylabel='Units Sold'>
```



What is the correlation between Units Sold, Sales, COGS, and Profit?

```
[141]: import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10,10))
correlation = df.corr(numeric_only=True)
sns.heatmap(
    correlation,
    annot=True,
    cmap='plasma',
    fct=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```



Explanation of the Correlation Heatmap Using Dataset

- 1) Gross Sales ↔ Sales ↔ COGS (≈ 0.99 – 1.00): Near-perfect correlation — expected, since Sales is derived from Gross Sales minus Discounts, and COGS scales directly with sales volume.
- 2) Gross Sales/Sales ↔ Profit (0.78 – 0.81): Strong positive relationship — higher sales generally drive higher profit, though not perfectly, since discounts and costs eat into margins.
- 3) Discounts ↔ Gross Sales/Sales (0.74 – 0.78): Higher discounts are associated with higher gross sales — likely because discounts are applied more on large-volume deals, not that discounting causes more sales.
- 4) Discounts ↔ Profit (0.38): Only a moderate relationship, weaker than Discounts ↔ Sales — suggesting heavy discounting boosts revenue but doesn't proportionally boost profit (it erodes margin).
- 5) Sale Price ↔ Profit (0.65): Moderately strong — pricing has a meaningful but not dominant effect on profitability.
- 6) Units Sold ↔ most variables (0.23 – 0.33): Surprisingly weak correlation with Sales/Profit — implying revenue is driven more by price and product mix than raw unit volume.
- 7) Manufacturing Price: Very weak correlation with almost everything (≈ 0.00 – 0.07) — manufacturing cost doesn't vary much with sales performance in this dataset.
- 8) Year: Near-zero correlation with all variables — no strong time-based trend captured here (makes sense, since correlation only checks linear year-over-year scaling, not seasonal patterns).

Create a regression model to predict sales

```
[150]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
df = pd.read_csv("Dataset.csv")

# Input and target
X = df[['Units Sold']]
y = df['Sales']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict sales
y_pred = model.predict(X_test)

# Check model performance
print("R2 Score:", r2_score(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

# Predict sales for a new quantity
new_quantity = [[10]]
prediction = model.predict(new_quantity)

print("Predicted Sales:", prediction[0])
```

```
R2 Score: 0.06535564527077464
Mean Squared Error: 51763524938.18301
Predicted Sales: 18040.775477365438
```

```
c:\users\krishnamohan b\appdata\local\programs\python\python39\lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
warnings.warn(
```

Excel

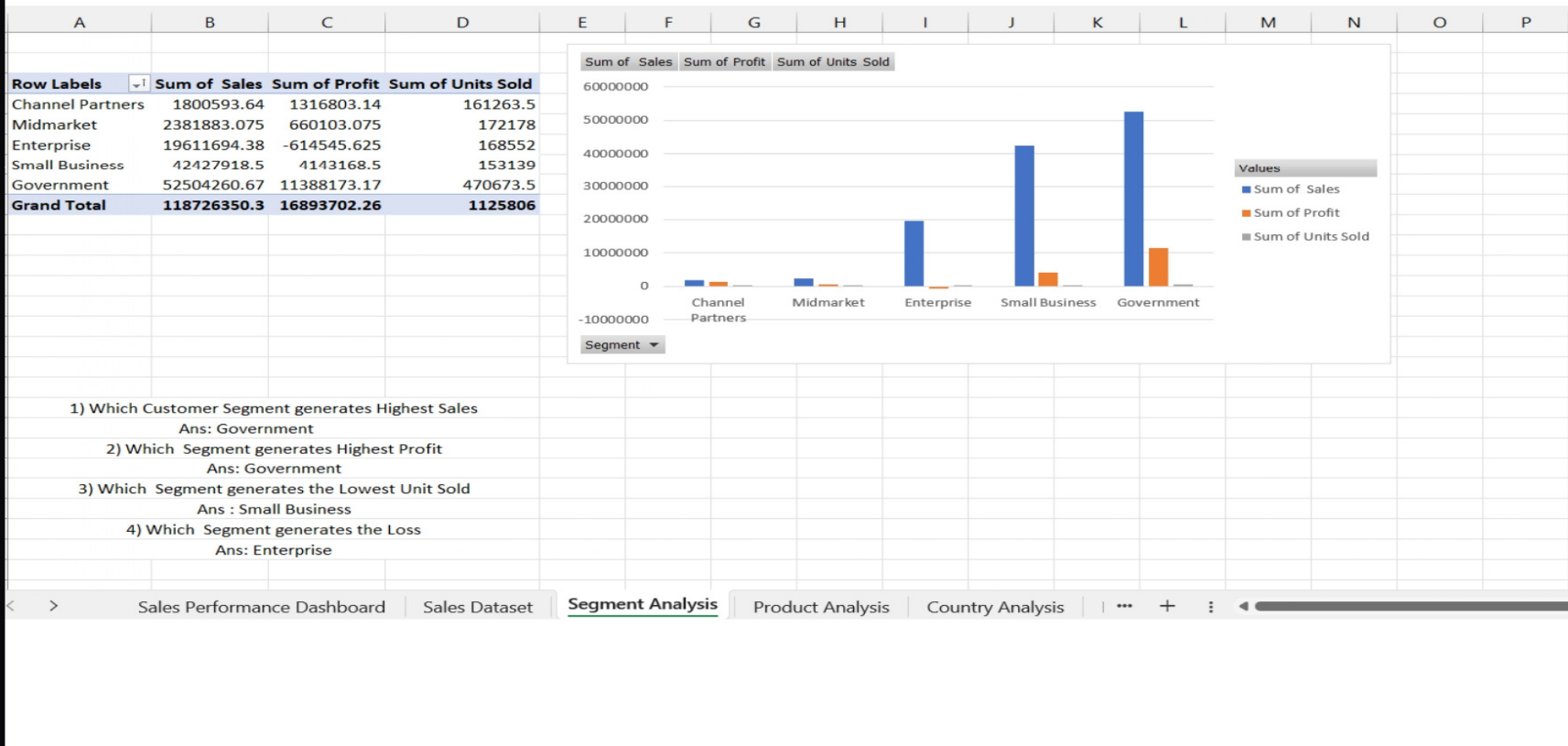
What is Excel ?

Microsoft Excel is a spreadsheet program developed by Microsoft that allows users to organize, calculate, format, and analyse data in a grid of rows and columns

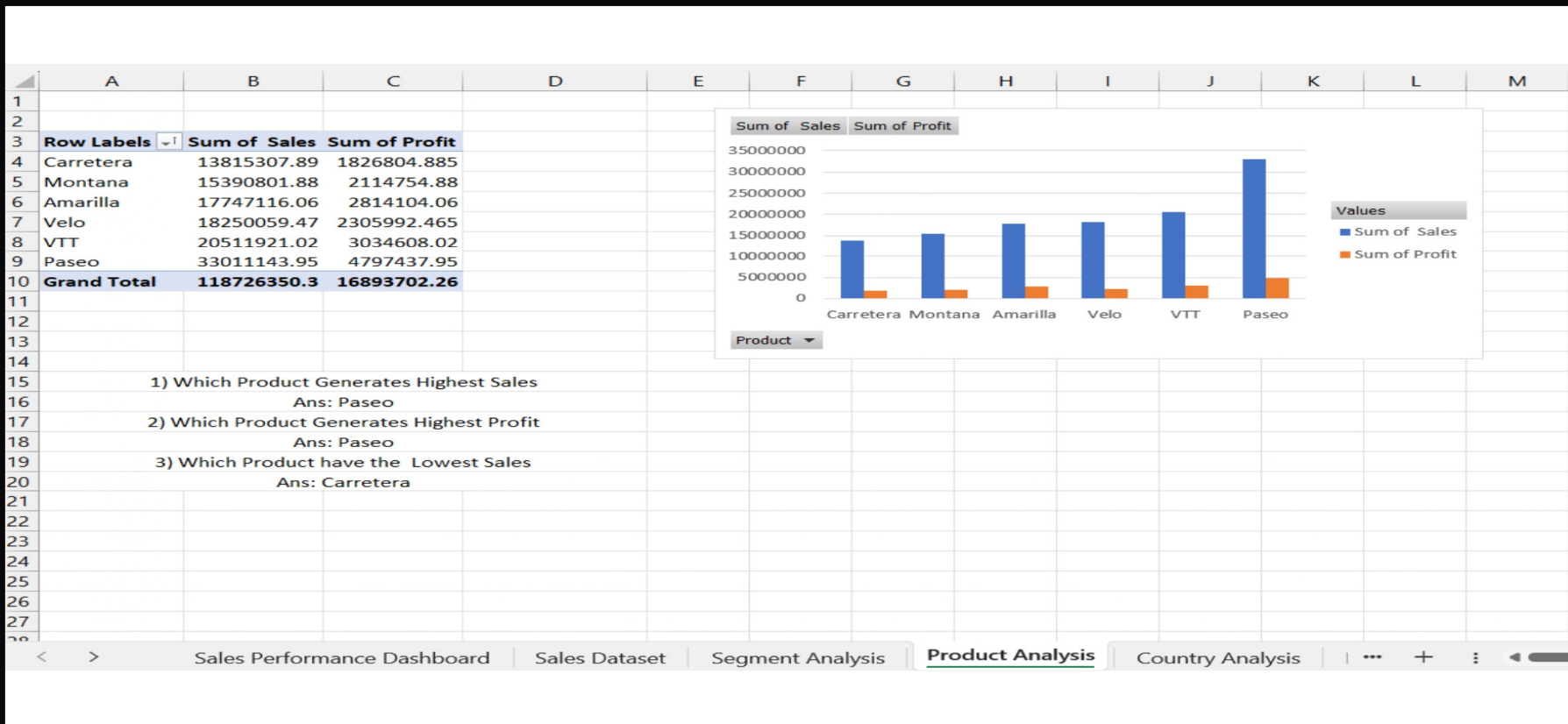
Sales Dataset (Using Excel)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Segment	Country	Product	Discount	Units Sold	Manufact	Sale Price	Gross Sale	Discounts	Sales	COGS	Profit	Date	Month Name	Year	Profit Margin		
Governme	Germany	Carretera	None	1513	3	350	529550	0	529550	393380	136170	01-Dec-14	December	2014	25.7142857		Total Sales
Governme	Germany	Paseo	None	1006	10	350	352100	0	352100	261560	90540	01-Jun-14	June	2014	25.7142857		118726350.3
Governme	Canada	Paseo	None	1725	10	350	603750	0	603750	448500	155250	01-Nov-13	November	2013	25.7142857		Total Profit
Governme	Germany	Paseo	None	1513	10	350	529550	0	529550	393380	136170	01-Dec-14	December	2014	25.7142857		16893702.26
Governme	Germany	Velo	None	1006	120	350	352100	0	352100	261560	90540	01-Jun-14	June	2014	25.7142857		Total Units Sold
Governme	France	VTT	None	1527	250	350	534450	0	534450	397020	137430	01-Sep-13	September	2013	25.7142857		1125806
Governme	France	Amarilla	None	2750	260	350	962500	0	962500	715000	247500	01-Feb-14	February	2014	25.7142857		Total Discount
Governme	Mexico	Carretera	Low	1210	3	350	423500	4235	419265	314600	104665	01-Mar-14	March	2014	24.963925		9205248.24
Governme	Mexico	Carretera	Low	1397	3	350	488950	4889.5	484060.5	363220	120840.5	01-Oct-14	October	2014	24.963925		Average Sales
Governme	France	Carretera	Low	2155	3	350	754250	7542.5	746707.5	560300	186407.5	01-Dec-14	December	2014	24.963925		169609.0718
Governme	France	Paseo	Low	2155	10	350	754250	7542.5	746707.5	560300	186407.5	01-Dec-14	December	2014	24.963925		Overall Profit Margin
Governme	Canada	VTT	Low	943.5	250	350	330225	3302.25	326922.8	245310	81612.75	01-Apr-14	April	2014	24.963925		14.2291094
Governme	Mexico	VTT	Low	1397	250	350	488950	4889.5	484060.5	363220	120840.5	01-Oct-14	October	2014	24.963925		
Governme	Canada	Carretera	Low	2852	3	350	998200	19964	978236	741520	236716	01-Dec-14	December	2014	24.1982507		
Governme	Canada	Paseo	Low	2852	10	350	998200	19964	978236	741520	236716	01-Dec-14	December	2014	24.1982507		
Governme	Germany	Velo	Low	2966	120	350	1038100	20762	1017338	771160	246178	01-Oct-13	October	2013	24.1982507		
Governme	Germany	Velo	Low	2877	120	350	1006950	20139	986811	748020	238791	01-Oct-14	October	2014	24.1982507		
Governme	Germany	VTT	Low	2877	250	350	1006950	20139	986811	748020	238791	01-Oct-14	October	2014	24.1982507		
Governme	United Sta	VTT	Low	266	250	350	93100	1862	91238	69160	22078	01-Dec-13	December	2013	24.1982507		
Governme	Mexico	VTT	Low	1940	250	350	679000	13580	665420	504400	161020	01-Dec-13	December	2013	24.1982507		
Governme	Germany	Amarilla	Low	2966	260	350	1038100	20762	1017338	771160	246178	01-Oct-13	October	2013	24.1982507		
Governme	Germany	Montana	Low	1797	5	350	628950	18868.5	610081.5	467220	142861.5	01-Sep-13	September	2013	23.4167894		
Governme	Mexico	VTT	Low	1642	250	350	574700	17241	557459	426920	130539	01-Aug-14	August	2014	23.4167894		
Governme	United Sta	Carretera	Low	274	3	350	95900	3836	92064	71240	20824	01-Dec-14	December	2014	22.6190476		
Governme	United Sta	Paseo	Low	3450	10	350	1207500	48300	1159200	897000	262200	01-Jul-14	July	2014	22.6190476		
Governme	United Sta	Paseo	Low	274	10	350	95900	3836	92064	71240	20824	01-Dec-14	December	2014	22.6190476		
Governme	France	Velo	Low	3477	120	350	1207500	48300	1159200	897000	262200	01-Jul-14	July	2014	22.6190476		

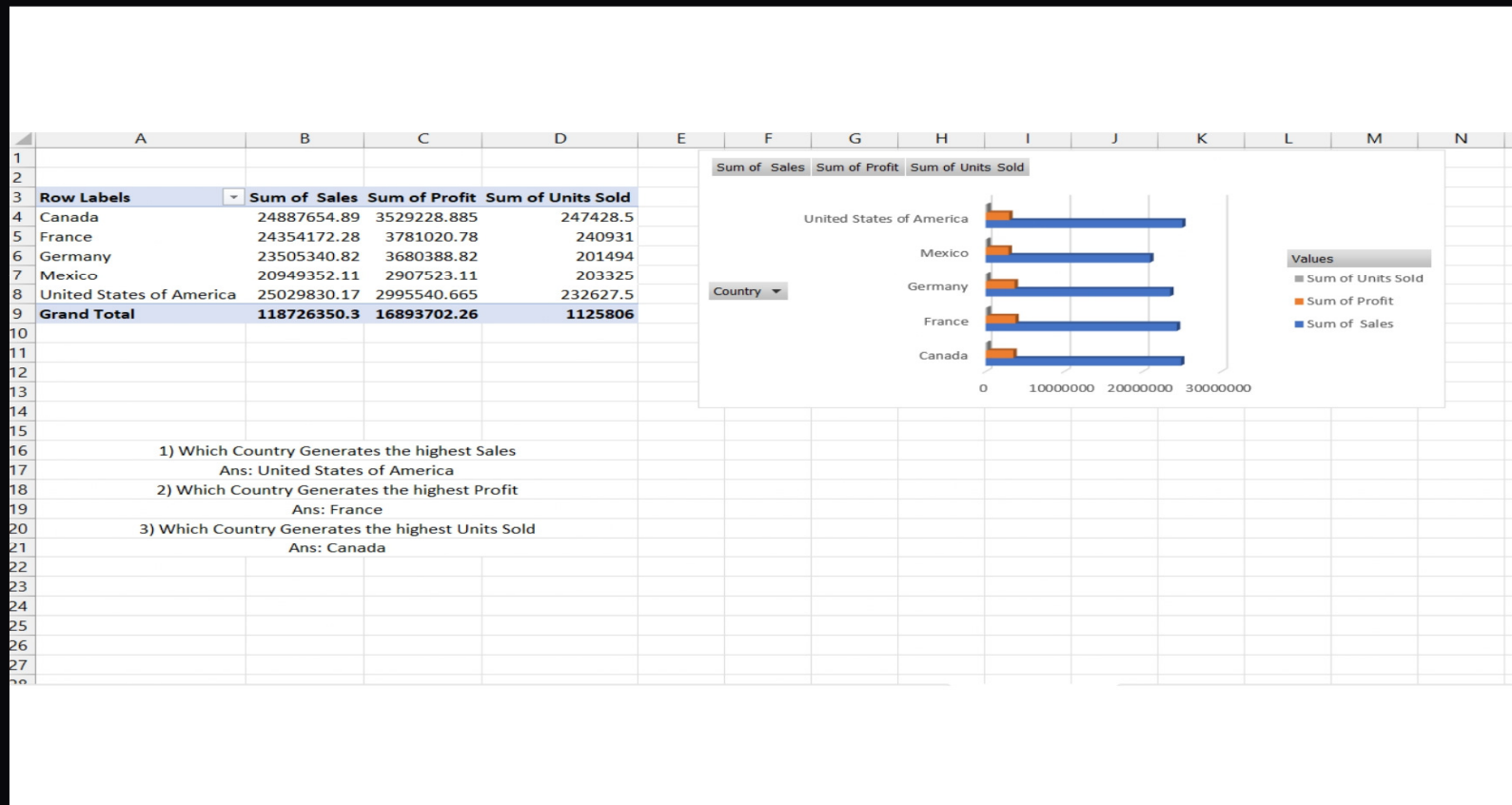
Segment Analysis



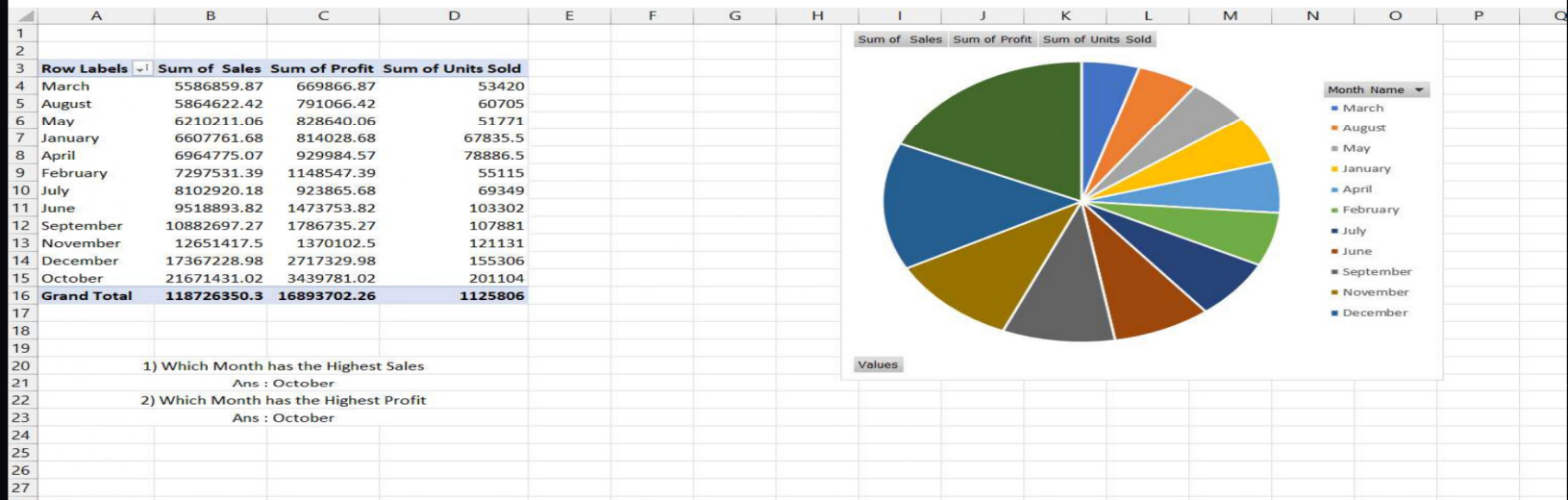
Product Analysis



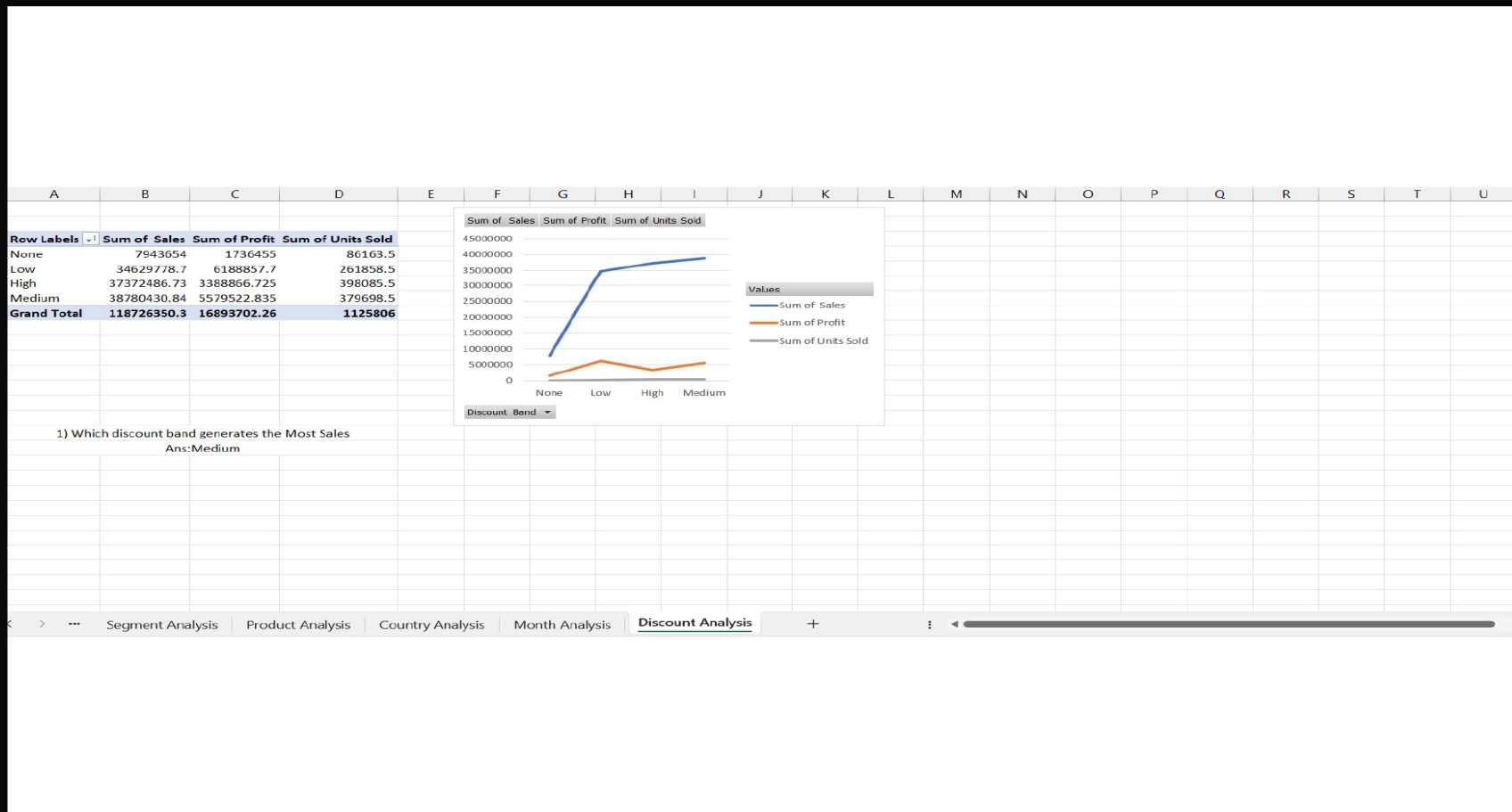
Country Analysis



Month Analysis



Discount Analysis



Power BI

What is Power BI ?

Power BI is Microsoft's business analytics and data visualization platform that turns raw, disconnected data into interactive, easy-to-understand dashboards and reports

Sales Performance Dashboard

Sum of sales
118.73M

Sum of Profit
16.89M

Average Sales
169.61K

Total no of Units sold
1.13M

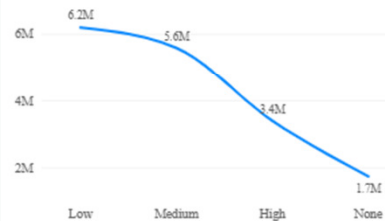
Total Discount
9.21M

Profit Margin %
14.23

Sum of Sales
by Year



Sum of Profit
by Discount Band



Month Name

April
August
December
February
January
July
June
March
May
November
October

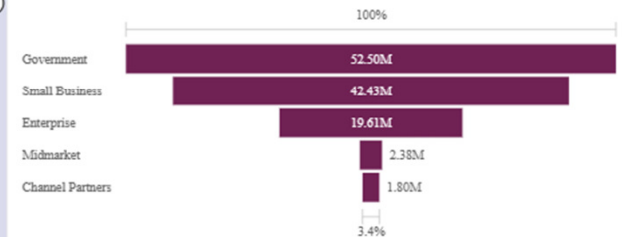
Year

2013 2014

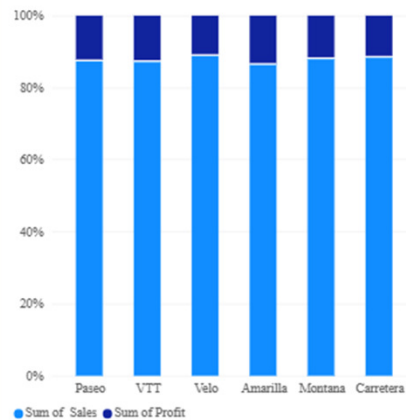
Product

☐ Amarilla
☐ Carretera
☐ Montana
☐ Paseo
☐ Velo
☐ VIT

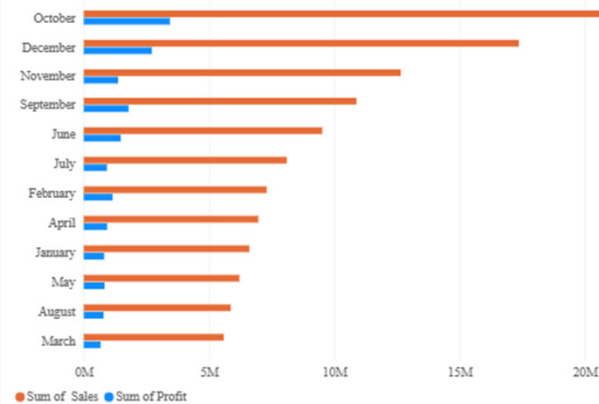
Sum of Sales
by Segment



Sum of Sales and Sum of Profit
by Product



Sum of Sales and Sum of Profit
by Month Name



Sum of Sales
by Country

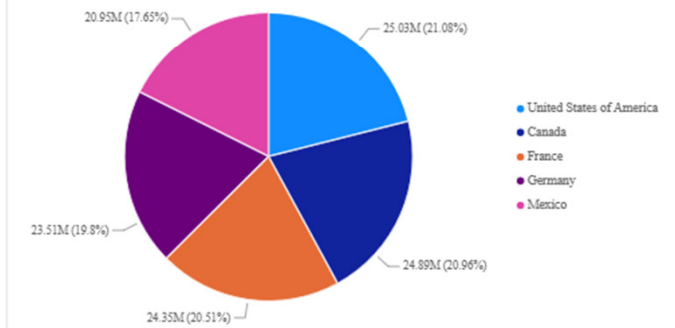


Table View

Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year
Government	Germany	Carretera	None	1513	3	350	529550	0	529550	393380	136170	01 December 2014	12	December	
Government	Germany	Paseo	None	1006	10	350	352100	0	352100	261560	90540	01 June 2014	6	June	
Government	Canada	Paseo	None	1725	10	350	603750	0	603750	448500	155250	01 November 2013	11	November	
Government	Germany	Paseo	None	1513	10	350	529550	0	529550	393380	136170	01 December 2014	12	December	
Government	Germany	Velo	None	1006	120	350	352100	0	352100	261560	90540	01 June 2014	6	June	
Government	France	VTT	None	1527	250	350	534450	0	534450	397020	137430	01 September 2013	9	September	
Government	France	Amarilla	None	2750	260	350	962500	0	962500	715000	247500	01 February 2014	2	February	
Government	Mexico	Carretera	Low	1210	3	350	423500	4235	419265	314600	104665	01 March 2014	3	March	
Government	Mexico	Carretera	Low	1397	3	350	488950	4889.5	484060.5	363220	120840.5	01 October 2014	10	October	
Government	France	Carretera	Low	2155	3	350	754250	7542.5	746707.5	560300	186407.5	01 December 2014	12	December	
Government	France	Paseo	Low	2155	10	350	754250	7542.5	746707.5	560300	186407.5	01 December 2014	12	December	
Government	Canada	VTT	Low	943.5	250	350	330225	3302.25	326922.75	245310	81612.75	01 April 2014	4	April	
Government	Mexico	VTT	Low	1397	250	350	488950	4889.5	484060.5	363220	120840.5	01 October 2014	10	October	Expand
Government	Canada	Carretera	Low	2852	3	350	998200	19964	978236	741520	236716	01 December 2014	12	December	
Government	Canada	Paseo	Low	2852	10	350	998200	19964	978236	741520	236716	01 December 2014	12	December	
Government	Germany	Velo	Low	2966	120	350	1038100	20762	1017338	771160	246178	01 October 2013	10	October	
Government	Germany	Velo	Low	2877	120	350	1006950	20139	986811	748020	238791	01 October 2014	10	October	
Government	Germany	VTT	Low	2877	250	350	1006950	20139	986811	748020	238791	01 October 2014	10	October	
Government	United States of America	VTT	Low	266	250	350	93100	1862	91238	69160	22078	01 December 2013	12	December	
Government	Mexico	VTT	Low	1940	250	350	679000	13580	665420	504400	161020	01 December 2013	12	December	
Government	Germany	Amarilla	Low	2966	260	350	1038100	20762	1017338	771160	246178	01 October 2013	10	October	
Government	Germany	Montana	Low	1797	5	350	628950	18868.5	610081.5	467220	142861.5	01 September 2013	9	September	
Government	Mexico	VTT	Low	1642	250	350	574700	17241	557459	426920	130539	01 August 2014	8	August	
Government	United States of America	Carretera	Low	274	3	350	95900	3836	92064	71240	20824	01 December 2014	12	December	
Government	United States of America	Paseo	Low	3450	10	350	1207500	48300	1159200	897000	262200	01 July 2014	7	July	
Government	United States of America	Paseo	Low	274	10	350	95900	3836	92064	71240	20824	01 December 2014	12	December	
Government	France	Velo	Low	2177	120	350	761950	30478	731472	566020	165452	01 October 2014	10	October	

Segment ▾	Sales ▾
Government	529550
Government	352100
Government	603750
Government	529550
Government	352100
Government	534450
Government	962500
Government	419265
Government	484060.5
Government	746707.5
Government	746707.5
Government	326922.75
Government	484060.5
Government	978236
Government	978236
Government	1017338
Government	986811
Government	986811
Government	91238
Government	665420
Government	1017338
Government	610081.5
Government	557459
Government	92064
Government	1159200
Government	92064
Government	731472

Country ▾	Sales ▾
Mexico	419265
Mexico	484060.5
Mexico	484060.5
Mexico	665420
Mexico	557459
Mexico	626640
Mexico	239400
Mexico	322420
Mexico	552391
Mexico	438564
Mexico	438564
Mexico	769814.5
Mexico	107156
Mexico	107156
Mexico	272888
Mexico	272888
Mexico	368676
Mexico	429660
Mexico	848172.5
Mexico	639922.5
Mexico	10662.4
Mexico	21112.4
Mexico	34112.4
Mexico	11327.4
Mexico	55071.2
Mexico	30830.8
Mexico	18891.6

Product ▾	Sales ▾
Paseo	352100
Paseo	603750
Paseo	529550
Paseo	746707.5
Paseo	978236
Paseo	1159200
Paseo	92064
Paseo	200165
Paseo	408310
Paseo	492184
Paseo	239183
Paseo	691012
Paseo	438564
Paseo	769814.5
Paseo	655551.75
Paseo	272888
Paseo	490952
Paseo	368676
Paseo	801444
Paseo	108706.5
Paseo	210700
Paseo	354277
Paseo	578522
Paseo	848172.5
Paseo	597082.5
Paseo	639922.5
Paseo	5840

DAX View

Sum of sales = SUM('Sales Dataset'[Sales])

Run Update model with changes (0)

```
1 EVALUATE
2   SUMMARIZECOLUMNS(
3     "Sum of sales", [Sum of sales]
4   )
```

Results Result 1 of 1 Copy

	[Sum of sales]
1	118726350.26

Sum of Profit = SUM('Sales Dataset'[Profit])

DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#)

Run

Update model with changes (0)

1

EVALUATE

2

SUMMARIZECOLUMNS

3

"Sum of Profit", [Sum of Profit]

4

Results

Result 1 of 1

Copy

	[Sum of Profit]
1	16893702.26

Total no of Units sold = sum('Sales Dataset'[Units Sold])

DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#)

Run

Update model with changes (0)

1

EVALUATE

2

SUMMARIZECOLUMNS

3

"Total no of Units sold", [Total no of Units sold]

4

Results

Result 1 of 1

Copy

	[Total no of Units so...
1	1125806

Average Sales = AVERAGE('Sales Dataset'[Sales])

DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#)

Run

Update model with changes (0)

1

EVALUATE

2

SUMMARIZECOLUMNS(

3

"Average Sales", [Average Sales]

4

)

Results

Result 1 of 1

Copy

	[Average Sales]
1	169609.07

Total Discount = SUM('Sales Dataset'[Discounts])

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 

  Update model with changes (0)

```
1 EVALUATE
2   SUMMARIZECOLUMNS(
3     "Total Discount", [Total Discount]
4   )
```

Results | Result 1 of 1  Copy 

	[Total Discount]
1	9205248.24

$$\text{Profit Margin \%} = [\text{Sum of Profit}] / [\text{Sum of sales}] * 100$$

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 

 Run  Update model with changes (0)

```
1 EVALUATE
2 SUMMARIZECOLUMNS(
3     "Profit Margin %", [Profit Margin %]
4 )
```

Results | Result 1 of 1  Copy 

	[Profit Margin %]
1	14.23

Summary of Findings and Recommendations

Summary of Findings

The analysis reveals some significant differences in terms of sales and profits by product, year, country, and customer segment. Certain products contribute much more towards total sales and profit than others. Units sold are shown to have an important relationship with sales, although there may be impacts from discounts affecting profit margins. Analysis at a year- and month-wise level allows us to see periods of increased sales performance.

Recommendations

High performing products/customer segments should focus efforts upon to increase sales & profitability as required. It would seem prudent to monitor levels of discounting in order to prevent unwarranted reductions in profit margins. It will be necessary to analyze low performing products and regions in search of avenues of improvements. Sales Trends could assist planning for inventories and marketing activities and sales strategy's for high demand times.

Thank You